

MD WAHIDUR RAHMAN

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• [LinkedIn Profile](#) • [ResearchGate Profile](#) • [Google Scholar](#)

RESEARCH INTEREST

- Seeking a research or engineering role in accelerator physics or beam dynamics, leveraging expertise in electron beam simulation, space charge modeling, and PIC analysis to advance next-generation particle accelerator systems.

SKILLS

- **Electromagnetic Simulation:** ECHO, Elegant, Mad-X
- **Particle-in-Cell (PIC), RF & Plasma Model Simulation:** XOOPIC, WarpX
- **RF & Plasma Modelling:** CST, COMSOL Multiphysics
- **Cleanroom Fabrication:** Photolithography, Thin Film Deposition, Plasma-Enhanced Chemical Vapor Deposition (PECVD), Physical Vapor Deposition (PVD): E-Beam Evaporation & Sputtering
- **Device Design Software:** Sentaurus TCAD, Mentor Graphics, Cadence, Synopsys VCS, Quartus (FPGA)
- **Circuit Design & Simulation:** Simulink, Multisim, LTspice, LabVIEW
- **Programming:** Assembly, C, Python, MATLAB, Mathcad, Mathematica, Arduino, CPLD

ACADEMIC BACKGROUND

Michigan State University (MSU), East Lansing, Michigan, USA Aug 2022 – May 2026
Thesis defended: Feb 2026; Degree awarded: May 2026
Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering GPA: 3.96
Thesis title: “Electron Beam-Wave Interaction in Linear Beam Devices.”

Michigan State University (MSU), East Lansing, Michigan, USA Aug 2022 – Aug 2025
Master of Science (M.S.) in Electrical and Computer Engineering GPA: 4.00

University of Technology Malaysia (UTM), Johor, Malaysia Feb 2018– Feb 2022
Bachelor of Engineering (B.Eng.) in Electrical and Electronic Engineering GPA: 3.86
Thesis title: “Fault Modelling of Gate-Oxide-Short in 15nm Junctionless FinFET.”

RESEARCH EXPERIENCES

Research Associate – Accelerator Physics July 2026 – Present
Electron-Ion Collider (Pre Operation)
Advisor: Alexei Blednykh (Project Manager)

Brookhaven National Laboratory, Upton, New York, USA

- Contribute to accelerator physics research for the Electron-Ion Collider (EIC).
- Perform advanced beam dynamics simulations to evaluate beam stability and accelerator performance.
- Collaborate with physics, engineering, and controls teams to support reliable collider operations.

Graduate Research Assistant, Plasma, Beams, and Interface Science Group Aug 2022 – May 2016

Michigan State University (MSU), Michigan, USA
Advisor: Prof. Peng Zhang (University of Michigan, Ann Arbor)
Co-advisor: Prof. John Verboncoeur (Michigan State University)

Project: Effect of Space Charge on Electron Beam-Wave Interaction

- Developed analytical models and conducted PIC (XOOPIC) simulations of single- and multi-cavity linear beam devices to analyse space charge effects, gap coupling factors, and beam dynamics.
- Research contributed to strategies for improving efficiency and stability of high-frequency RF amplifiers in particle acceleration and fusion applications.

Project: Gap Coupling Factor of an Annular Electron Beam

- Developed analytical and disk models with space charge effects and performed PIC/CST simulations to study annular beam–gap dynamics under large-signal conditions.

- Demonstrated enhanced coupling strength and efficiency over solid beams, advancing the design of high-power microwave devices.

Project: Discrete Cavity Analysis of Traveling-Wave Tube (TWT)

- Designed a coupled-cavity TWT amplifier using Pierce's theory, modelling cavities as coupled oscillators and applying eigen-analysis to study beam-circuit interaction, gain and cavity tunability..
- Improved efficiency and validated the discrete model, overcoming challenges in modelling high-frequency tubes, providing insights for optimized amplifier design.

Project: Fabrication and Simulation (Class 100 Cleanroom)

- Fabricated FinFET devices on p-type silicon wafers using E-beam PVD, photolithography, and thin-film deposition in a Class 100 cleanroom.
- Validated fabrication process and device performance through Sentaurus TCAD simulation and experimental characterization, ensuring process reliability.

Undergraduate Research Assistant

Aug 2020 – Feb 2022

University of Technology Malaysia (UTM), Johor, Malaysia

Supervisor: Prof. Nurul Ezaila Alias

Project: Fault Modelling of Gate-Oxide-Short (GOS) in Junctionless FinFET

- Designed and characterized 15 nm junctionless n- and p-channel FinFETs in Sentaurus, developing a gate-oxide-short (GOS) fault model with pinhole defects.
- Analysed performance shifts and delay variations, providing insights for circuit-level reliability of nanoscale devices.

PROFESSIONAL EXPERIENCES

Research Assistant (Intern), Computational Nanoelectronics (CoNE)

May 2021 – Aug

2021

University of Technology Malaysia, Johor, Malaysia

Project: Defect Analysis of Multibridge Channel MOSFET (MBCFET)

- Modelled and simulated n- and p-channel MBCFET-based inverter circuits in Sentaurus TCAD to investigate the effects of device-level defects on circuit operation.
- Analysed defect-induced variations in switching delay, and threshold voltage, improving nanoscale circuit reliability.

Research Assistant (Intern), DHL, Malaysia

Sep 2020 – Jan 2021

Project: Smart Automatic Parcel Sorting System

- Developed a machine vision-based barcode scanning system using real-time image processing and interval-based scanning for faster parcel sorting.
- Improved sorting accuracy and operational efficiency by reducing manual scanning and handling errors.

PUBLICATIONS

PEER REVIEWED JOURNALS

- [1] **M. W. Rahman**, P. Wong, J. Luginsland, M. Franzi, B. Hoff, D. Simon, N. Jordan, R. Gilgenbach, and P. Zhang, "Discrete Cavity Analysis of Coupled-Cavity Traveling Wave Tubes," under review, 2026.
- [2] **M. W. Rahman**, P. Zhang, "Gap Coupling Factor of an Annular Electron Beam," IEEE Trans. Plasma Sci., vol. 53, no. 10, pp. 2933–2938, Oct. 2025.
- [3] **M. W. Rahman**, P. Zhang, "Parametric analysis of electron beam-wave interaction in linear beam devices: A tutorial on gap coupling factor and scaling to high frequency," AIP Adv., vol. 15, no. 3, p. 035128, Mar. 2025.

CONFERENCE ARTICLES

- [1] **M. W. Rahman**, P. Wong, J. Luginsland, M. Franzi, B. Hoff, D. Simon, N. Jordan, R. Gilgenbach, and P. Zhang, "A Discrete Cavity Analysis for Coupled-Cavity Traveling Wave Tubes," 2026 IEEE International Conference on Plasma Science (ICOPS) (June 22-26, 2026, Lake Tahoe, Nevada, USA).
- [2] **M.W. Rahman**, P. Zhang, "The Effect of Space Charge on the Performance of Linear Beam Devices", 2024 IEEE Vacuum Electronics Conference (IVEC) (April 22-25, 2024, Monterey, California, USA).
- [3] **M.W. Rahman**, P. Zhang, "Space Charge Effects on Beam-Wave Interaction at the Output Gap of Vacuum Electron Tube", 2023 Gaseous Electronics Conference (GEC) (October 9-13, 2023, Ann Arbor, Michigan, USA).
- [4] **M.W. Rahman**, P. Zhang, "Parametric Analysis of Electron Beam-Cavity Interaction", 2023 IEEE International Conference on Plasma Science (ICOPS) (May 21-25, 2023, Santa Fe, New Mexico, USA).

- [5] **M.W. Rahman**, N.E. Alias, A. Hamzah, M.P. Tan, I. Kamisian, "Comprehensive Analysis of Gate Oxide Short in Junctionless Fin Field Effect Transistor," 2022 IEEE International Conference on Semiconductor Electronics (ICSE), pp. 73-76, August 2022.

POSTER PRESENTATIONS

- [1] **M.W. Rahman**, P. Zhang, "Effects of Annular Beam Properties on Gap Coupling in High-Frequency Microwave Devices", 2024 Michigan Institute for Plasma Science & Engineering (MIPSE) (December 3, 2025, Ann Arbor, Michigan, USA).
- [2] **M.W. Rahman**, P. Zhang, "Parametric Analysis of the Beam Kinetic Energy in Linear Beam Devices", 2024 Michigan Institute for Plasma Science & Engineering (MIPSE) (November 20, 2024, Ann Arbor, Michigan, USA).
- [3] **M.W. Rahman**, P. Zhang, "Space Charge Effect on Electron Beam-Cavity Interaction", 2024 US Low Temperature Plasma Summer School (USLTPSS) (June 27, 2024, Ann Arbor, Michigan, USA).
- [4] **M.W. Rahman**, P. Zhang, "The Effect of Space Charge on the Performance of Linear Beam Device for High Frequency Radio Waves", 2023 Michigan Institute for Plasma Science & Engineering (MIPSE) (November 15, 2023, Ann Arbor, Michigan, USA).
- [5] **M.W. Rahman**, P. Zhang, "Large Signal Analysis for Space Charge Effect on Electron Beam-Cavity Interaction", 2023 American Vacuum Society (AVS) Michigan Chapter (June 7, 2023, Ann Arbor, Michigan, USA).

INVITED TALK & PEER REVIEW ACTIVITY

- [1] [Invited Talk] **M.W. Rahman**, "Gap Coupling Factor for Solid and Annular Electron Beam Devices," presented at the AFOSR Multidisciplinary University Research Initiative (MURI) Teleseminar, hosted by the Air Force Office of Scientific Research (AFOSR), June 2025.
- [2] [Peer Review] **M.W. Rahman**, Reviewer for the 2025 2nd International Conference on Electronic and Computer Engineering (ECE2025), March 2025.

GRADUATE CERTIFICATES

- **Accelerator Science and Engineering Traineeship (ASET)**, Physics Astronomy, MSU, USA 2025
- **Semiconductor Manufacturing, Processing & Devices**, ECE, MSU, USA 2024

AWARDS & ACHIEVEMENTS

- **Graduate Office Fellowship (GOF)**, ECE, Michigan State University, Michigan, USA 2026
- **PhD Dissertation Fellowship (DCF)**, Michigan State University, Michigan, USA 2025
- **MSU Fellowship Grant**, Michigan State University, Michigan, USA 2023 – 2024
- **Gold Medal & Dean's Award**, UTM, Malaysia 2022
- **Dean's List Fellowship**, Electrical Engineering, UTM, Malaysia 2018 – 2022
- **Education Board Scholarship**, Dhaka, Bangladesh 2008 – 2016

VOLUNTARY WORKS

- **NPSS MSU**, Michigan State University, Michigan, USA Aug 2022 – Aug 2025
Served as Vice President at IEEE Nuclear and Plasma Science Society (NPSS) Student Chapter at MSU.
- **Community Writers**, University of Technology Malaysia Aug 2019 – Dec 2019
Taught Math and English to underprivileged students in Sagil Village National School, Johor, Malaysia.

ADDITIONAL INFORMATION

- **Visa Status:** F-1 OPT; eligible for STEM OPT — no employer visa sponsorship required.